

Effect of removing polypropylene fibre surface finishes on mechanical performance of kenaf/polypropylene composites

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ABSTRACT

Natural fibre/polypropylene thermoplastic composites are often produced by compression moulding of a blended preform of polypropylene fibre and natural fibre treated by chemicals or enzymes. Two preform processing routes may be adopted: (1) treating the natural fibre first and then blending it with the polypropylene fibre (the pre-treatment route), and (2) forming a blended preform of the natural fibre and polypropylene fibre first and then carrying out the chemical/enzyme treatment on the blended preform (the post-treatment route). The kenaf/polypropylene composites produced according to the post-treatment route show up to 36% higher flexural strength and up to 63% higher flexural modulus than the composites produced according to the corresponding pre-treatment route. These differences were attributed to the chemical surface finishes of the polypropylene fibre, which have been removed in the post-treatment processing route, but persisted into the final composites in the pre-treatment processing route.

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1. Introduction

Natural fibres such as flax, hemp, kenaf and sisal are increasingly used as replacements for glass and other non-renewable fibres in composite materials mainly because they are abundant, cheap and renewable. Their other advantages include low density, high toughness, comparable specific strength properties, reduction in tool wear, low energy consumption in fabrication, and CO₂ neutrality [1]. The main drawback in using these natural fibres is their hydrophilic nature which generally leads to problems of adhesion with hydrophobic polymer matrices.

Much of the work in natural fibre composites has dealt with the effects of coupling agents on the mechanical properties of natural fibre reinforced thermoplastics such as polypropylene (PP) with considerable emphasis on chemical modification for tailoring the properties of natural fibres [2,3]. The use of a coupling agent such as maleated polypropylene (MAPP) obtained by grafting maleic anhydride (MA) onto the PP polymer chains has been shown to be very effective [4,5]. The use of MAPP for the improvement of mechanical properties of composites with thermoplastic matrices especially with PP matrix reinforced with various natural fibres (e.g., hemp, kenaf, flax and wood) has been extensively studied and well understood [4–10]. Other approaches for improving natural fibre composite performances that have been widely studied include separating elementary fibres from their bundles, removing

non-cellulosic compounds and modifying the fibre surface morphology [11–21].

The most commonly used method to remove non-cellulosic materials from plant cellulose fibres is an alkali treatment that uses sodium hydroxide (NaOH) to remove pectin, lignin, and water- and alkali-soluble hemicelluloses [12]. This causes the bundle fibres to split into elementary fibres. The process is known as degumming in the textile industry for softening flax fabrics in apparel and household applications. Ethyl alcohol has been used to extract waxy substances from cotton fibres [13]. The chelating agent EDTA (ethylene diamine tetra-acetic acid) and its phosphonated analogue EDTMPA (ethylene diamine tetra methylene phosphonic acid) have been used to remove calcium ions from pectin in natural fibre cell walls to make pectin soluble in liquids [14,15] and to separate elementary fibres from their bundles. EDTMPA can strongly interact with mineral surfaces and can be easily removed from natural systems with a lower impact on the environment than EDTA. Non-cellulosic constituents of natural fibres can also be removed by using enzymes [16,17]. For example, pectate lyase can degrade pectin, and laccase along with a mediator can depolymerise lignin [16]. However, waxes and other non-cellulosic compounds like dusty materials and hemicelluloses can be a barrier to these enzymes. In order to achieve good results, these enzyme treatments should be preceded by a hot alcohol–water mixture to remove waxy substances, water soluble hemicelluloses and dusts [18–21].

A widely adopted approach for the fabrication of natural fibre reinforced polypropylene thermoplastic composite automotive parts involves first producing a blended preform of natural fibre and thermoplastic fibre and then tailoring and hot-pressing the

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blended preform to form the required composite parts [22]. Commercial thermoplastic fibres such as polypropylene are commonly treated with surface finishing agents to facilitate textile processing. Fibre finishing agents are complex mixtures of ingredients such as lubricants, emulsifiers, antistatic agents, wetting agents, and antimicrobial compounds. For example, paraffin waxes and mineral oils are used as finishing agents to reduce friction between the fibres and machine parts while retaining a suitable cohesion between the fibres during textile processes [23]. These finishing agents are removed from the fibre after the textile processing is completed, for example, during fabric desizing, dyeing and wet finishing. Because natural fibre/PP preforms for composite fabrication do not require dyeing and wet finishing, any surface finishes left on the PP fibre will persist into the final composites.

In our recent study of chemical and enzymatic treatments of kenaf fibre for composite manufacture, we noticed a consistent improvement of composite performance related to the sequence of fibre processing and treatments performed on the kenaf and PP fibres: (1) chemically or enzymatically treating the kenaf fibre first and then mixing it with untreated PP fibre to form a preform (pre-treatment route), and (2) mixing the kenaf and PP fibres to form a preform first and then subjecting the blended preform to chemical or enzymatic treatments (post-treatment route). The key difference between the two processing routes is that the pre-treatment route does not treat the PP fibres (i.e., the surface finishes on PP fibre persist into final composites) while the post-treatment route removes the PP fibre surface finishes before forming final composites. Composites made from the post-treatment route showed consistently better mechanical performance than those from the pre-treatment route.

2. Experimental

2.1. Materials

Kenaf fibres were obtained from mechanical breaking of kenaf plant at source in Malaysia. Tensile testing was performed on untreated technical fibres (bundles of elementary fibres) according to ASTM D3822-07. The testing was carried out using an INSTRON 5500R fitted with a 100 N-load cell. The gauge length was 20 mm. The fibre specimen was weighed in order to determine the linear density (tex or mg/m). The average cross-sectional area of the specimen was then calculated from the fibre linear density and fibre volumetric density. The reported volumetric density of kenaf fibre varies over a wide range from 1.2 g/cm³ [10] to 1.4 g/cm³ [11]. We adopted the average value of 1.3 g/cm³ in this study. About 50 technical fibres were tested to obtain the average tensile properties.

The tensile properties of technical kenaf fibres were determined as follows: breaking stress of 433.1 MPa (standard deviation 186.0 MPa), Young's modulus of 26.9 GPa (standard deviation 8.1 GPa) and elongation at break of 1.8% (standard deviation 0.57%).

Polypropylene fibre in loose bale form (mean fibre length of 101 mm, 15 denier) normally used for carpet yarn spinning was obtained through a carpet company in Australia. Scourzyme[®] L (pectate lyase, activity: 375 APSU-CA/g), laccase (NS 29044, activity 800 LAMU/g) along with a mediator (NS 29045) were obtained from Novozyme Australia Pty Ltd. EDTMPA (chelating agent) was obtained from BIO-RAD. The ethanol, sodium hydroxide, adipic acid, potassium dihydrogen orthophosphate and petroleum spirits used in this study were all of analytical grade.

2.2. Mechanical processes

The kenaf fibres were first passed through a Fearnought opener twice to obtain relatively clean kenaf fibres. The cleaned kenaf fibres were further processed by the following two routes.

2.2.1. Pre-treatment route

Kenaf fibre was first treated according to the procedures described in Section 2.3. An equal weight of polypropylene fibre was then mixed with the treated kenaf fibre. The kenaf/PP fibre mixture of 50/50 by weight was carded together on an Eroel Wheels Mini Card to achieve an intimately blended web that has a preferential orientation along the machine direction. The fibre web accumulated on the card was then carefully removed by hand. Three layers of such webs were laid in the same direction to form a mat of about 450 g/m².

2.2.2. Post-treatment route

The untreated kenaf fibre and polypropylene fibre in equal weight were mixed and carded on the Eroel Wheels Mini Card. The intimately mixed and preferentially aligned blended fibre webs removed from the card were laid in the same direction to form a mat in the same way as in the pre-treatment route. The blended kenaf/PP mat was then subjected to chemical and enzymatic treatments described in Section 2.3.

2.3. Chemical and enzymatic treatments

Materials for treatment (kenaf fibre, PP fibre or kenaf/PP blended preform) were first dried at 80 °C for 12 h. The materials were then placed into stainless steel canisters fitted with magnetic stirrers for chemical and enzyme treatments. The treatments were carried out using an Ahiba Turbomat laboratory dyeing machine set to the required temperature. The procedures for these treatments are described in Table 1. A sample to liquor ratio of 20–1 was used for all treatments except the laccase treatment (d) where a 10–1 ratio was used. Following pectate lyase (b), chelator (c) and laccase (d) treatments, fibres were washed with a mixture of 0.05 wt.% Dekol SNS and 0.05 wt.% Sandopan MRN (Sandoz, Clarisant) at 80 °C for 20 min. After the completion of all chemical and enzyme treatments, fibres were washed in hot boiling water for about 10 min and then in cold water for an additional 10 min. The fibres were then dried at room temperature overnight in a ventilated fume cupboard.

2.4. Fabrication of kenaf/PP composites and neat PP sheets

The kenaf/PP preforms were dried at 80 °C for 12 h before hot pressing. The preforms were cut along the principal fibre direction (i.e., the longitudinal direction) and the transverse direction in order to fit into an aluminium mould and then placed in a preheated press for 15 min at a temperature of 190 °C under a constant pressure of 0.6 MPa. The mould was then placed in a cold press to cool down under a pressure of 0.2 MPa. The resultant composite sheets were approximately 3 mm in thickness. Pure polypropylene fibre mats, both untreated and treated, were also made into neat polypropylene sheets in the same way.

Composite samples that had no visible defects were used for testing. There were no apparent differences between the composite samples as a result of the different chemical treatments applied to the fibres. A representative specimen from each composite sample was used to determine composite density. A theoretical density was calculated based on the component material ratio of 50/50 by weight between kenaf and PP, which was used as basis for the calculation of the void contents (i.e., porosity) of the composite samples.

2.5. Flexural testing

Composite sheets and neat PP sheets were cut into flexural test specimens for three point bend testing. Flexural testing was carried out in accordance with ASTM D 790-03 using an Instron 5500R

Table 1

Procedures for the different chemical and enzyme treatments.

Treatment	Description
(a) Ethanol/water	Fibres were treated in a mixture of 50 vol.% ethanol and 50 vol.% water at 80 °C for 2 h [21–23].
(b) Pectate lyase	Fibre was pre-treated with (a) and then treated with 0.5 wt.% aqueous solution of Scourzyme L (pH adjusted to 8.5 with 0.1 N NaOH) at 60 °C for 6 h as per recommendation by the supplier.
(c) Chelator	Fibre was pre-treated with (a) and then treated with 0.5 wt.% aqueous solution of EDTMPA (pH adjusted to 11 with 0.1 N NaOH) at 60 °C for 6 h as per recommendation by the supplier.
(d) Laccase	Fibre was pre-treated with (a) and then treated with a mixture containing 15 wt.% laccase, 7 wt.% mediator, 39 wt.% potassium dihydrogen orthophosphate and 39 wt.% adipic acid (pH adjusted to 4.5 with 0.1 N NaOH) at 65 °C for 1 h as per recommendation by the supplier.
(e) Alkali	Fibre was treated in 5 wt.% aqueous solution of NaOH at 120 °C for 60 min [14].

tensile testing machine. Six replicates were tested for each sample. The statistical comparison of mean values was carried out at the 95% confidence limit using a two-sample Student's *t* test.

2.6. Fibre diameter measurement

The diameters of untreated and treated kenaf fibres were measured using an OFDA100 Optical Fibre Diameter Analyser [24]. The instrument magnifies and captures images of individual fibres using a video camera and the diameter of each fibre is automatically determined by image analysis. OFDA tests were carried out on three randomly chosen specimens from each fibre sample, giving a total of about 6000 measurements which were pooled together to calculate a mean diameter and a standard deviation.

2.7. Fibre surface finish analysis

The surface finish present on the PP fibres was investigated by attenuated total reflectance (ATR) infrared spectroscopy. Spectra were recorded using a Perkin Elmer Spectrum 100 Fourier transform spectrometer fitted with a diamond/ZnSe single bounce micro-ATR accessory and a deuterated triglycine sulphate detector. A total of 128 scans were collected at a resolution of 2 cm⁻¹ for each spectrum. All samples were run in triplicate. Fibre spectra were normalised on the 1454 cm⁻¹ CH₂ deformation vibration which can almost completely be associated with polypropylene. To better characterise the fibre surface finish, untreated PP fibre was washed using warm (40 °C) petroleum spirits and the solvent was allowed to partially evaporate, concentrating the residue. The spectrum of the residue was obtained from a film cast on a KBr plate and analysed in transmission mode.

3. Results and discussion

3.1. Effectiveness of kenaf fibre treatments

Fig. 1 shows the flexural strength (top) and modulus (bottom) along the principal fibre direction of the composites produced from the untreated and treated kenaf/PP preforms according to the pre-treatment route. The results obtained from untreated neat PP sheets are also included for comparison. Compared to the neat PP sheets, the average flexural strength of all composite samples were between 60% and 84% higher and their flexural moduli were between 180% and 284% higher. These results indicate that the kenaf fibre provided good reinforcement to the PP matrix. Fig. 1 also shows that in comparison with the composite produced from the untreated kenaf/PP preform, the chemical and enzyme treatments on kenaf fibre in general did not significantly improve the average flexural properties of composites in the principal fibre direction. The only significant improvement over the untreated sample was in the flexural modulus of the alkali treated sample which showed a 22% increase.

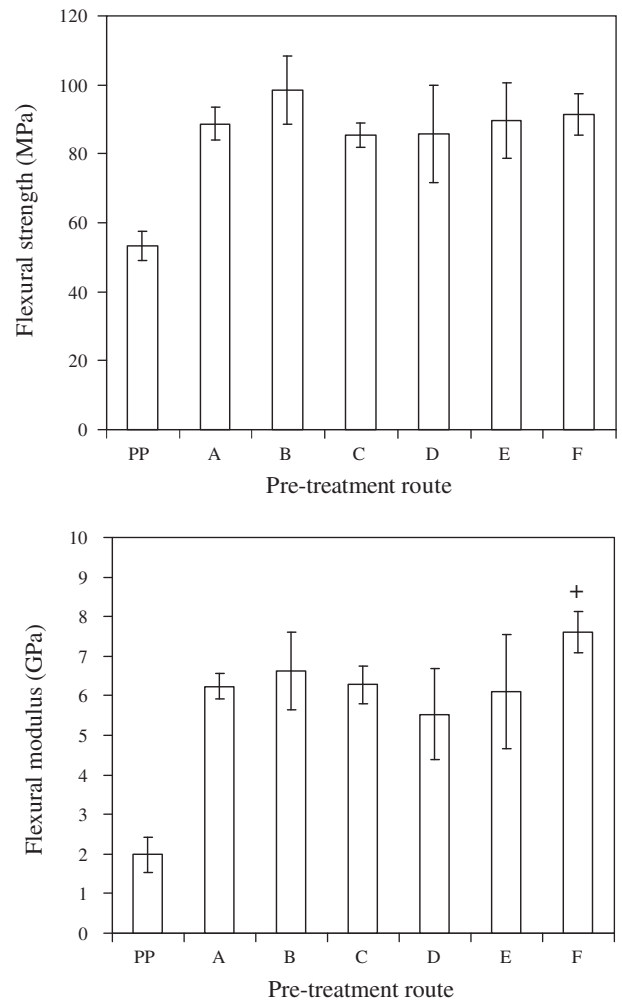


Fig. 1. Flexural properties in principal fibre direction of composites produced according to the pre-treatment route: (PP) neat PP, (A) untreated, (B) ethanol/water, (C) pectate lyase, (D) cheator, (E) laccase and (F) alkali treatments. + Significantly different from the untreated sample.

Table 2 presents the flexural strength and modulus along the transverse fibre direction of the composites. The largest differences between the kenaf reinforced and un-reinforced (i.e., neat PP) samples were about 14% and 95% for flexural strength and modulus, respectively. These are substantially smaller than the increases in the corresponding longitudinal properties. The flexural strength of all the treated kenaf/PP composites in the transverse direction (Table 2) was not significantly different from the untreated kenaf/PP composite. The transverse flexural modulus of three treated kenaf/PP composites (ethanol/water, chelator and alkali) was found to be significantly different from the untreated kenaf/PP

Table 2

Flexural properties in transverse direction of composites produced from untreated and pre-treated preforms.

Treatment	Strength (MPa)	Modulus (GPa)
Untreated neat PP	53.4 (4.3)	1.98 (0.44)
Untreated kenaf/PP	58.5 (4.1)	3.50 (0.25)
Ethanol/water	60.1 (9.9)	2.78 (0.41)
Pectate lyase	54.7 (3.7) ^Δ	3.03 (0.55) [×]
Chelator	60.7 (8.0)	2.86 (0.23)
Laccase	60.6 (12.9)	3.86 (0.83) ^Δ
Alkali	60.2 (7.3)	2.94 (0.41)

Standard deviation in bracket.

^Δ Not statistically different from the untreated kenaf/PP sample.

[×] Not quite statistically different from the untreated kenaf/PP sample.

composite while the laccase treated sample was not significantly different and the pectate lyase treated sample was not quite significantly different from the untreated kenaf/PP sample.

3.2. Effect of preform processing routes on composite flexural properties

Fig. 2 compares the flexural strength and modulus in the principal fibre direction of composites produced according to the post-treatment route with the composites produced by the pre-treatment route. The post-treatment route resulted in significantly higher flexural strength and modulus than the corresponding com-

posites produced according to the pre-treatment route irrespective of the treatment methods used. The only exceptions were the flexural strength of ethanol/water treated composite and the flexural modulus of alkali treated composite, which showed no statistically significant difference between the two treatment routes at 95% level of confidence. However, if the statistical test is relaxed slightly, both of these samples show small improvements. The improvements range from 10% to 36% in flexural strength and from 8% to 63% in flexural modulus. The chelator treatment provided the highest improvement in both flexural strength (36%) and flexural modulus (63%).

The porosities of the composite samples are presented in Table 3. Although there are variations between individual samples, the average porosity of the pre-treatment composites was very close to that of the post-treatment composites, indicating that there was no systematic difference between the two processing routes.

As it will be discussed in Section 3.5, chemical treatments of kenaf fibres lead to the dissolving (or degumming) of some non-cellulose substances in the kenaf fibre, resulting in a small change in the kenaf fibre content of the final composites. In the pre-treatment route, the treated kenaf fibre was mixed with PP fibre at a 50/50 weight ratio, while in the post-treatment route, untreated kenaf fibre and PP fibre mixed at a 50/50 weight ratio was treated. Consequently the kenaf fibre content in the final composites manufactured through the pre-treatment route should be slightly higher than that in the corresponding composites manufactured through the post-treatment route. As higher fibre content generally leads to improved mechanical performance of the final composites according to the rule of mixtures, the benefit of the post-treatment route over the pre-treatment route shown in Fig. 2 would be somewhat greater if the comparisons were carried out on the basis of equal kenaf fibre content.

In contrast, the composites produced by the two processing routes demonstrated no statistically significant difference in flexural strength and flexural modulus in the transverse direction for all treatment methods (Table 3).

3.3. Effect of treatments on neat polypropylene plastic sheets

Polypropylene fibres were subjected to the same treatments as listed in Table 1 and then hot-pressed into neat polypropylene plastic sheets. The flexural properties of the neat plastic sheets are presented in Table 4. The chemical and enzymatic treatments of the PP fibre increased the average flexural strength of the PP sheets by between 4% and 9%, and the flexural modulus by between 1.5% and 8%, although not statistically significant. Polypropylene fibre has high chemical resistance [25]. The chemical and enzymatic treatments in Table 1 should only affect the surface finish and have little if any effect on the fibre itself. The moderate increases in flexural properties observed for the neat PP plastics prepared from the treated fibre may be attributable to the removal of PP fibre surface finish. If not removed, the surface finish could become a source of structural defects in the plastic sheets and composites.

3.4. Effect of treatments on polypropylene fibre surface finish

Surface finishes can constitute a considerable portion of the fibre material [23,26,27]. For carpet yarns, typical applications range from 0.7% to 1.2% of the weight of the yarn [27,28]. Finishing agents are complex materials and their exact compositions are usually not provided by their manufacturers. The effect of the various chemical and enzyme treatments on the surface finish present on the PP fibres was investigated by ATR infrared spectroscopy. The low wavenumber spectrum obtained from the untreated PP fibres as trace A in Fig. 3. The weak peaks at 1114 cm^{-1} (ether C—O—C

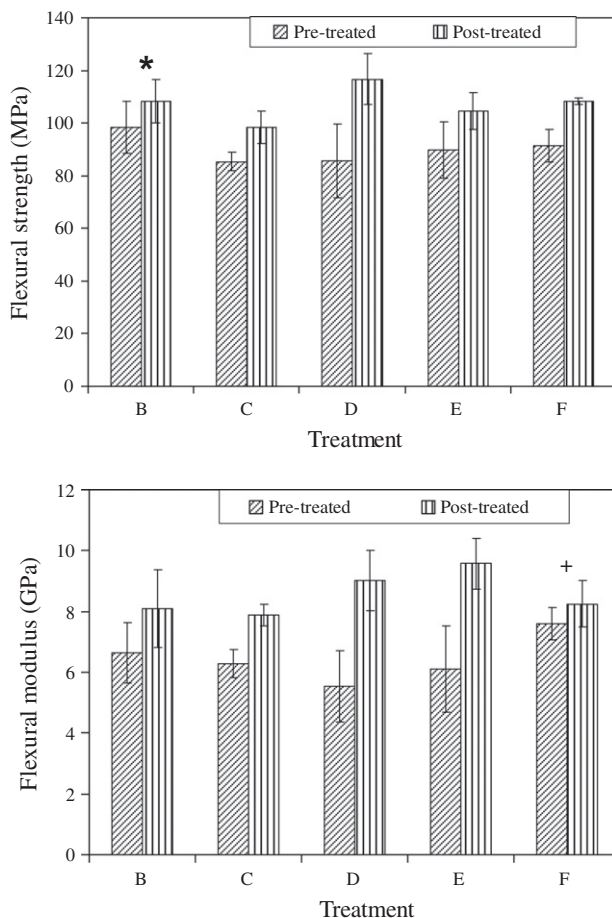


Fig. 2. Influence of treatment route on flexural properties in principal fibre direction of kenaf/PP composites: (B) ethanol/water, (C) pectate lyase, (D) cheator, (E) laccase and (F) alkali treatments. * Not quite significantly different, + Not significantly different.

Table 3

Porosity and flexural properties in transverse direction of composites produced according to pre-treatment and post-treatment routes.

Treatment	Porosity		Strength (MPa)		Modulus (GPa)	
	Pre-treatment	Post-treatment	Pre-treatment	Post-treatment	Pre-treatment	Post-treatment
Untreated	0.060		58.5 (4.1)		3.50 (0.25)	
Ethanol/water	0.062	0.045	60.1 (9.9)	59.0 (4.1)	2.78 (0.41)	2.78 (0.58)
Pectate lyase	0.028	0.064	54.8 (3.7)	54.9 (6.7)	3.03 (0.56)	2.80 (0.56)
Chelator	0.014	0.046	60.8 (8.0)	59.8 (4.3)	2.86 (0.24)	2.92 (0.49)
Laccase	0.030	0.020	60.6 (12.9)	61.8 (9.3)	3.86 (0.83)	3.59 (0.82)
Alkali	0.065	0.038	60.2 (7.3)	62.4 (2.3)	2.94 (0.42)	2.95 (0.25)
Average	0.040	0.043	59.3	59.6	3.09	3.01

Standard deviation in bracket.

Table 4

Flexural properties of untreated and treated neat PP plastic sheets.

Treatment	Strength (MPa)	Modulus (GPa)
Untreated	53.4 (4.3)	1.98 (0.44)
Ethanol/water	57.2 (7.2)	2.02 (0.32)
Pectate lyase	56.6 (5.2)	2.01 (0.53)
Chelator	58.2 (8.6)	2.10 (0.37)
Laccase	57.4 (5.6)	2.08 (0.50)
Alkali	55.7 (4.4)	2.14 (0.45)

Standard deviation in bracket.

asymmetric stretching vibration [29]) and 1738 cm^{-1} ($\text{C}=\text{O}$ ester stretching vibration [29]) are not typical of polypropylene and thus can be attributed to the fibre surface finish. Relative to the PP bands, the intensity of the 1114 cm^{-1} band appears to vary between replicate spectra by more than 50% while that of the 1738 cm^{-1} band varies by only 36%. Further, when the ether band is at its maximum the ester band was found to be at a minimum, suggesting that there are at least two components present in the finish.

The ATR spectra obtained from the chemical and enzyme treated PP fibres are shown as traces B through F of Fig. 3. The 1114 cm^{-1} ether band is no longer observed, suggesting that all of the treatments are efficient at removal of this component. In contrast, the carbonyl band appears to persist, suggesting that this component is more substantive on the fibre surface. After ethanol/water (B), alkali (F) and laccase (E) treatments, this component appears to increase in intensity. Careful examination revealed that

this increase is actually due to the presence of a new band component at 1744 cm^{-1} with the 1738 cm^{-1} band still present as a weak shoulder. This new component cannot be associated with residual Dekol SNS or Sandopan MRN as they were not used in the ethanol/water treatment. Spectral bands near 1745 cm^{-1} are generally associated with free carboxylic acids [29] which could be produced by partial hydrolysis of esters present in the fibre finish.

Petroleum spirits (PS) can be used to remove surface finishes from PP fibres. The spectra obtained from the surface of the PP fibres before and after rinsing with PS are shown in Fig. 4 as traces A and PS respectively. PS rinsing had a comparable effect to that of the wet treatments in removing the ether component of the surface finish (1114 cm^{-1} feature). It also had a greater effect on removal of ester functionality while producing no carboxylic acid groups.

The infrared spectrum obtained from the residue left after evaporation of the PS used to rinse the untreated PP fibres is shown as trace A of Fig. 5. When the same amount of solvent used to rinse the fibres was allowed to evaporate, no residue was detected. The spectrum obtained from the rinse residue suggests that the fibre finish is made up largely of two functional groups, a polyethylene oxide (PEO) component (1114 cm^{-1}) and an ester component (1737 cm^{-1}). These features are consistent with those identified through the ATR surface analysis of the untreated fibres. Strong aliphatic C–H stretching vibrations are also observed near 2900 cm^{-1} [29]. While the band components attributable to CH_2 groups dominated, the highest frequency component (2952 cm^{-1}) is indicative of CH_3 groups [29], possibly due to low molecular weight PP extracted by the PS. The spectrum obtained from the extract residue

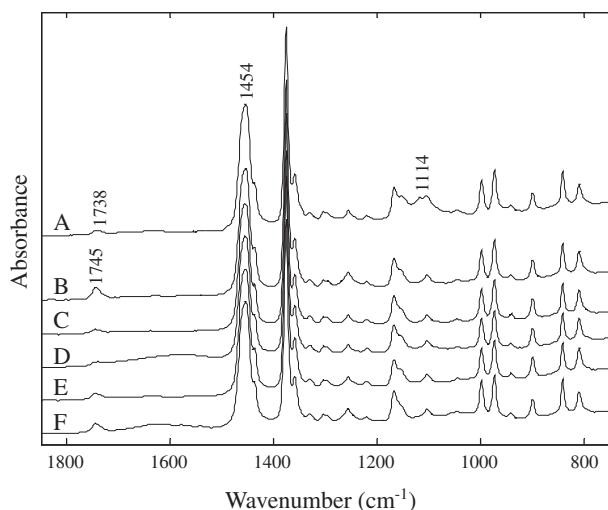


Fig. 3. Low wavenumber region of the infrared ATR spectra obtained from (A) untreated, (B) ethanol/water, (C) pectate lyase, (D) cheator, (E) laccase and (F) alkali treated PP fibres.

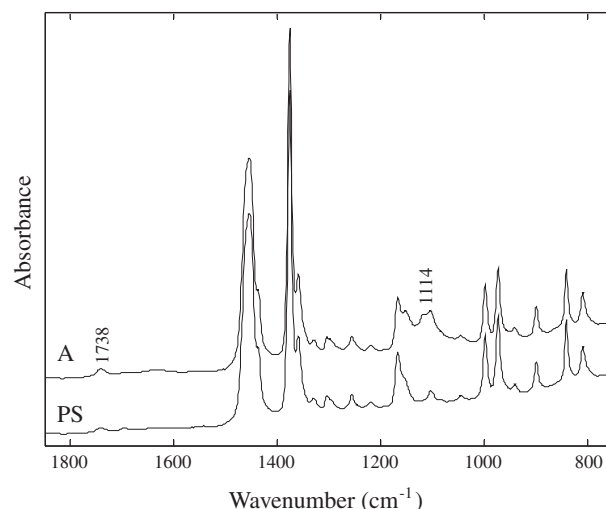


Fig. 4. Low wavenumber region of the infrared ATR spectra obtained from (A) untreated and (PS) petroleum spirits washed PP fibres.